

Todo list

Measuring Diversity and Segregation with Convex Functions

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We consider the problem of *segregation measurement*. Colloquially, a population of individuals is segregated with respect to a given trait if individuals with different values of that trait tend to be separated from one another by geography, policy, or other social forces. The segregation measurement problem is to construct compact, quantitative descriptions of the extent and structure of segregation in a given population. The purpose of such a measurement is often comparative—we aim to say that one population (say, a city) is more segregated, or differently segregated, than another. We’ll focus here on the problem of *spatial* segregation, in which individuals are assumed to be located space (often at their place of residence) and we seek to understand how the strongly the spatial distribution of individuals reflects their demographic traits.

Figure 1 provides a classical illustration, as well as a few of its complexities. In each of the four panels, individuals with demographic possible demographic traits A or B are arranged into a synthetic “city.” Panel (a) describes such a city in which only members of group B are present; intuitively such a city cannot display spatial sorting between groups because there is only one group present. Our first intuitive insight is then that *diversity is necessary for segregation*. Panel (b) shows that diversity, however, is not sufficient; if each group is represented uniformly across the map, then there is also no spatial segregation. Panels (c) and (d) do display spatial segregation, with each individual location containing a uniform population of either 100% A or 100% B . Since each panel contains the same number of locations dominated by each group, any measure of segregation which “forgets” the spatial structure of the population will assign

the same value to both panels; such a measure may be considered *aspatial*. The problem of constructing *spatial* measures (which successfully distinguish between panels (c) and (d) in intuitive ways) has often been called the “checkerboard problem” in the quantitative sociology literature [RO04]. To summarize, a *diversity* measure can distinguish panels (a) and (b); an *aspatial segregation* measure can distinguish both from panel (c), and a *spatial* segregation measure can distinguish (d) from (c).

Any spatial measure which does successfully distinguish (c) and (d), however, is also subject to the *modifiable areal unit problem (MAUP)* [Ope84]: the method of aggregating continuous geographic space into discrete areal units (e.g. counties, tracts, blockgroups, or neighborhoods) is often arbitrary and can have a strong effect on the resulting segregation measurement. By way of illustration, if in panel (c) the resolution of our measurement was coarsened to the scale of 16×16 blocks, illustrated by the dark box, then we would be unable to distinguish (c) from (b).

The study of quantitative measures for spatial segregation extends far enough back that a review article in 1955 could already express frustration and disappointment with the chaotic state of the methodology. Commenting then on the measures extant at the time, Duncan and Duncan [DD55] wrote:

1. *The proposed indexes of segregation have a number of hitherto unnoticed interrelationships which can be mathematically demonstrated.*
2. *Some of them have mathematical properties of which their proponents were unaware, and which lead to difficulties of interpretation.*
3. *As a consequence, the status of the empirical work already done with segregation indices is questionable,*

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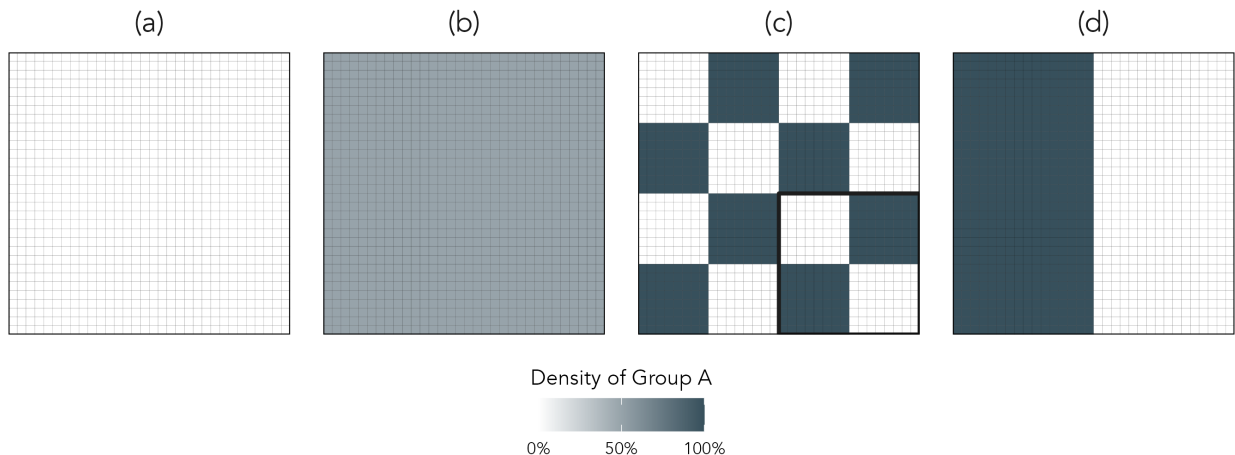


Figure 1: Synthetic “cities” composed of 32×32 spatial units. Each contains an equal density of population, and individuals within the population may be of either group A (black) or group B (white). Panel (c) is further arranged into 4 blocks of 8 cells each.

and their validity for further research is undermined.

Since that time, there have been efforts within quantitative sociology to impose a more unified mathematical structure on segregation measures, especially in the work of Sean Reardon and collaborators [RF02, Rea, ?RO04]. Still, over 70 years later, hints of the chaos described by Duncan and Duncan remain. This author is aware of two instances within the last ten years of a diversity or segregation measure being published as novel despite already existing in the literature under a different name. Somewhat distinctly from the sociological literature, there has been some recent attention on segregation measurement in the mathematics community [DMW23, TBK⁺25, HSS25]. These works, though interesting and important in their own right, tend to be somewhat disconnected from recent progress within the sociology community.

In this article, I argue that beginning with some of the systematic work in quantitative sociology not only enriches opportunities for conversation between mathematicians and social scientists but also provides a rich ground for interesting, accessible mathematics and data analysis. I do not attempt to offer a thorough review of prior work on segregation measurement, but rather of-

fer my particular perspective and the sources which have informed it. After establishing notation, we begin with a discussion of diversity measures, arguing that standard proposed axioms for these measures lead naturally to the class of convex functions defined on the probability simplex. Comparing global and average local diversities leads directly to a segregation measure interpretable as a measure of the information content that the spatial location carries about demographic information. We give several proofs showing that this general information measure satisfies many of the desirable properties of segregation measures proposed in the quantitative sociology literature. We then demonstrate two ways to “spatialize” this information, one using conditional information decomposition and another using the locally quadratic structure of Bregman divergences to construct a local measure of the spatial scale of segregation. We close with some discussion about extending this framework in the direction of ordinal segregation measures, as well as some open mathematical challenges.

1 Preliminaries

We first define abstract notions of “populations” and “population measurements.” Our framework is intentionally discrete, in recognition of the fact that all Census-type data sets provide demographic information in discrete areal units such as counties, tracts, or block-groups.

Definition 1. For $\ell \in \mathbb{N}$, the *probability simplex* Δ_ℓ is the set

$$\Delta_\ell = \left\{ \mathbf{y} \in \mathbb{R}^\ell : \sum_{j=1}^{\ell} y_j = 1, y_j \geq 0 \text{ for all } j \in [\ell] \right\}.$$

of probability vectors on ℓ elements.¹

Definition 2. A *population* $\mathcal{P}$ with n locations in d spatial dimensions and ℓ demographic categories is a triple $\mathcal{P} = (\mathbf{X}, \mathbf{Y}, \mu)$, where:

- $\mathbf{X} \in \mathbb{R}^{n \times d}$ is a matrix with rows $\mathbf{x}_i$, $i = 1, \dots, n$.²
- $\mathbf{Y} \in \mathbb{R}^{n \times \ell}$ is a row-stochastic matrix: for each row i , $\mathbf{y}_i \in \Delta_\ell$.
- $\mu \in \Delta_n$.

We denote the set of all populations with n locations, d dimensions, and ℓ demographic categories as $\mathcal{P}_{nd\ell}$.

In this definition, the matrix $\mathbf{X}$ represents space: element $\mathbf{x}_i$ of $\mathbf{X}$ can be viewed as giving the location of the i th spatial subunit of the population. The matrix $\mathbf{Y}$ represents the demographic distribution of the population according to some attribute such as gender, race, or education level: the i th row $\mathbf{y}_i$ of $\mathbf{Y}$ gives the proportion of individuals in each demographic category at location i . Finally, the probability vector μ represents the population density across locations; the proportion of individuals living in location i is μ_i . The probability that a single individual, chosen uniformly at random from the population, lives in location i and is in demographic category

j is $\mu_i y_{ij}$. The mean population vector $\bar{\mathbf{y}} = \sum_{i=1}^n \mu_i \mathbf{y}_i$ has entries giving the marginal probabilities that an individual chosen uniformly at random from the entire population belongs to each demographic category.

Definition 3. A *population measurement* is a function $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$.

We consider the problems of quantifying diversity, segregation, and spatial segregation as the problems of constructing population measurements which match intuition and satisfy desirable properties.

2 Diversity and Convexity

Heuristically, we often say that a population is *diverse* with respect to an attribute when it contains many categories of that attribute in roughly equal proportions. Diversity is usually considered to be a property of a population as a whole, without regard for spatial or demographic subdivisions. Thus, a diversity measure $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$ is a function of the mean population demographic vector: $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ for some function $\psi : \Delta_\ell \rightarrow \mathbb{R}$. What properties should this function have in order to capture an intuitive notion of diversity?

Working in mathematical ecology, Pielou [Pie77] gives a proposed set of criteria. For $\ell \in \mathbb{N}$, let $[\ell] = \{1, 2, \dots, \ell\}$. Let $\mathbf{e}_I \in \Delta_\ell$ be the vector with 1 in the indices specified by the multi-index $I \subseteq [\ell]$ and 0 elsewhere.

1. **Uniform maximizer.** “Diversity should be maximized when all groups are present in equal proportions.” Formally, $\frac{1}{\ell} \mathbf{e}_{[\ell]} = \arg\max_{\mathbf{y} \in \Delta_\ell} \psi(\mathbf{y})$.
2. **Monotone under group addition.** “In two populations in which all groups are equally represented, the population with the greatest number of groups will be more diverse.” Formally, if $I \subset [\ell]$, then $\psi(\mathbf{e}_I / |I|) < \psi(\mathbf{e}_{[\ell]} / \ell)$.

We will soon find that properties (1) and (2) are satisfied by a broad range of functions ψ . Pielou also offers a third axiom, which is more restrictive and essentially characterizes a single diversity measure; we discuss it in Appendix A.1.

¹Although it is sometimes common to refer to Δ_ℓ as an $(\ell - 1)$ -simplex (this being its manifold dimension), we write Δ_ℓ to emphasize its role as the set of probability distributions over ℓ categories.

² $\mathbf{X}$ can be replaced by a discrete metric space $(\mathcal{X}, d)$ where d describes measurement of distance between locations; under this replacement all the remaining results in this work hold except for ????

Pielou’s proposed axioms capture important intuitions, but they don’t offer a recipe for constructing useful diversity functions ψ . A hint is offered by the group addition property; we can think of a population uniformly distributed across ℓ demographic categories as a weighted mixture of two populations, one with $|I|$ categories and another with $\ell - |I|$ categories. One candidate generalization of the group addition property is therefore that *diversity should increase on average when populations are mixed*. To formalize, let $\mathbf{y}_1$ and $\mathbf{y}_2$ be two population demographic vectors and let $\lambda \in (0, 1)$ be a mixture weight (expressing, for example, the relative population sizes). Mixing the populations we can form a new demographic vector $\mathbf{y} = \lambda \mathbf{y}_1 + (1 - \lambda) \mathbf{y}_2$. Our mixing intuition suggests that if $\mathbf{y}_1 \neq \mathbf{y}_2$ then

$$\psi(\mathbf{y}) > \lambda \psi(\mathbf{y}_1) + (1 - \lambda) \psi(\mathbf{y}_2). \quad (1)$$

Here, the lefthand side gives the diversity of the mixed population $\mathbf{y}$, while the righthand side gives the λ -weighted average diversity of the original populations $\mathbf{y}_1$ and $\mathbf{y}_2$.

Definition 4. Set $S \in \mathbb{R}^d$ is a *convex set* if for all $\mathbf{y}_1, \mathbf{y}_2 \in S$ and $\lambda \in [0, 1]$, we have $\lambda \mathbf{y}_1 + (1 - \lambda) \mathbf{y}_2 \in S$. If S is a convex set, then $\phi : S \rightarrow \mathbb{R}$ is a *convex function* if for all $\mathbf{y}_1, \mathbf{y}_2 \in S$ and $\lambda \in [0, 1]$, we have $\phi(\lambda \mathbf{y}_1 + (1 - \lambda) \mathbf{y}_2) \leq \lambda \phi(\mathbf{y}_1) + (1 - \lambda) \phi(\mathbf{y}_2)$. Function ϕ is *strictly convex* if the above inequality is strict whenever $\mathbf{y}_1 \neq \mathbf{y}_2$ and $\lambda \in (0, 1)$. Function ψ is (strictly) *concave* if $\phi = -\psi$ is (strictly) convex.

Throughout the remainder of this article, we state without proof the convexity of certain functions as well as general properties of convex functions. I recommend [BV23] for a thorough treatment and proofs.

The simplex Δ_ℓ is a convex set, and so the notion of convexity applies to functions defined on Δ_ℓ . We will abuse notation slightly and refer to a population measurement $f(\mathcal{P}) = -\phi(\bar{\mathbf{y}})$ as (strictly) *concave* if ϕ is (strictly) convex.³ Equation (1) says that ψ should be a strictly concave function.

Example 2.1 (Shannon Entropy). If $\phi_H(\mathbf{y}) = \sum_{j=1}^\ell y_j \log y_j$, the resulting diversity measurement

³Following convention, we will emphasize convexity of ϕ rather than concavity of $-\phi$.

$\psi_H = -\phi_H$ is the *Shannon entropy*, which is widely used in ecology, sociology, and information theory.

Example 2.2 (Simpson Index). If $\phi_S(\mathbf{y}) = \sum_{j=1}^\ell y_j^2 - 1$, the resulting diversity measurement $\psi_S = -\phi_S$ is the *Simpson index*, which is widely used in ecology and sociology. The Simpson index has an attractive physical interpretation: it is the probability that two individuals chosen uniformly at random from the population belong to different demographic categories.

Example 2.3 (A Bespoke Economic Equality Measurement). For each $w \in 0, \dots, w_{\max}$, define a demographic category as “possessing exactly w units of wealth.” Then, entry y_{iw} gives the proportion of individuals in location i possessing wealth w . Let $\mathbf{C}$ be the lower triangular matrix of ones: $\mathbf{C}_{ij} = 1$ if $i \geq j$ and $\mathbf{C}_{ij} = 0$ otherwise. Then $\mathbf{C}\mathbf{y}_i$ gives the cumulative distribution function of wealth at location i . Fix a target mean wealth $w \in \mathbb{N}$. The function $\phi_{\text{eq}}(\mathbf{y}) = \|\mathbf{C}(\mathbf{y} - \mathbf{e}_w)\|_2^2$ is strictly convex, and so $f(\mathcal{P}) = -\|\mathbf{C}(\bar{\mathbf{y}} - \mathbf{e}_w)\|_2^2$ is a concave population measurement. This population measurement expresses a value on wealth equality; it is maximized when $\bar{\mathbf{y}} = \mathbf{e}_w$, i.e. when all individuals in the population possess exactly w units of wealth.

Remark 2.1. By definition, the economic equality measure of Example 2.3 depends on the choice of target mean wealth w . As we will see later (Equation (3)), segregation measures based on this diversity measure are independent of w .

Theorem 2.1. If $\phi : \Delta_\ell \rightarrow \mathbb{R}$ is strictly convex, then the population measurement $f(\mathcal{P}) = -\phi(\bar{\mathbf{y}})$ satisfies Pielou’s monotonicity under group addition property.

Definition 5. Function ϕ is *symmetric* if for all $\mathbf{y} \in \Delta_\ell$ and all permutations σ of $[\ell]$, we have $\phi(\mathbf{y}) = \phi(\sigma(\mathbf{y}))$, where $\sigma(\mathbf{y})$ is the vector obtained by permuting the entries of $\mathbf{y}$ according to σ . Function $\phi : \Delta_\ell \rightarrow \mathbb{R}$ is *separable* if there exists a function $g : [0, 1] \rightarrow \mathbb{R}$ such that $\phi(\mathbf{y}) = \sum_{j=1}^\ell g(y_j)$ for all $\mathbf{y} \in \Delta_\ell$.

Remark 2.2. Separability implies symmetry, but the converse is not true. Both the Shannon entropy ψ_H and the Simpson index ψ_S are separable (and therefore symmetric), while the economic equality measure ψ_{eq} is neither.

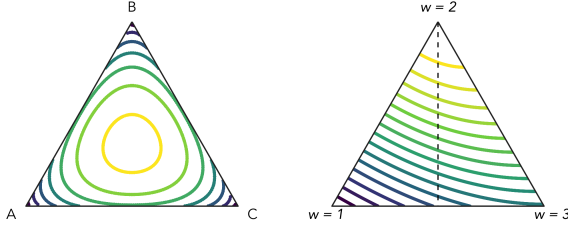


Figure 2: (Left): Isocontours of the Shannon entropy on the simplex Δ_3 . (Right): Isocontours of the economic equality measurement function from Example 2.3 with possible wealth values $w \in \{1, 2, 3\}$ and mean wealth $\bar{w} = 2$. The subset of Δ_3 corresponding to distributions with mean wealth $\bar{w} = 2$ is given by the dashed line.

Theorem 2.2. *If $\phi : \Delta_\ell \rightarrow \mathbb{R}$ is strictly convex and symmetric, then the population measurement $f(\mathcal{P}) = -\phi(\bar{\mathbf{y}})$ satisfies Pielou’s uniform maximizer property.*

Remark 2.3. The Shannon and Simpson diversity measures are both strictly convex and symmetric, and therefore satisfy both properties. The equality measure of Example 2.3 is strictly convex but not symmetric, and therefore satisfies the monotonicity under group addition property but not the uniform maximizer property.

3 Segregation and Information

Examining panels (b-d) of Figure 1, we see that in panel (b) each location is exactly as diverse as the population as a whole ($\phi(\mathbf{y}_i) = \phi(\mathbf{y})$ for each i), while in panels (c) and (d) each location is less diverse than the population as a whole. This suggests a simple quantitative segregation measure: the difference between global and mean local diversities.

Definition 6. Given a concave diversity function $f = -\phi$, the *Jensen information* [BGW05] of a population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$ is

$$I_\phi(\mathcal{P}) = f(\bar{\mathbf{y}}) - \sum_{i=1}^n \mu_i f(\mathbf{y}_i) = \sum_{i=1}^n \mu_i [\phi(\mathbf{y}_i) - \phi(\bar{\mathbf{y}})] . \quad (2)$$

Jensen’s Inequality, a standard result in convex analysis, implies that $I_\phi(\mathcal{P}) \geq 0$ for all populations $\mathcal{P}$ and convex functions ϕ . If ϕ is strictly convex, then $I_\phi(\mathcal{P}) = 0$ if and only if $\mathbf{y}_i = \bar{\mathbf{y}}$ for all $i \in [n]$, i.e. every location has the same demographic distribution as the population as a whole.

Although the Jensen information eq. (2) is nonnegative, individual terms of the form $\phi(\mathbf{y}_i) - \phi(\bar{\mathbf{y}})$ can be negative. This means that a given location’s “contribution to segregation” cannot simply be interpretably measured from the terms in eq. (2). It would instead be convenient for analytic purposes to write the Jensen information as a sum of nonnegative contributions from each location:

$$I_\phi(\mathcal{P}) = \sum_{i=1}^n \mu_i g(\mathbf{y}_i, \bar{\mathbf{y}}) , \quad (3)$$

Theorem 3.1. *Let $\phi : \Delta_\ell \rightarrow \mathbb{R}$ be a strictly convex function which is differentiable on $\text{relint}(\Delta_\ell)$. Then, eq. (3) holds for some function $g : \text{relint}(\Delta_\ell) \times \Delta_\ell \rightarrow \mathbb{R}$ and for any population $\mathcal{P} \in \mathcal{P}_{nd\ell}$ if and only if g is the Bregman divergence induced by ϕ :*

$$g(\mathbf{y}, \bar{\mathbf{y}}) = D_\phi(\mathbf{y} \parallel \bar{\mathbf{y}}) = \phi(\mathbf{y}) - \phi(\bar{\mathbf{y}}) - \nabla \phi(\bar{\mathbf{y}})^\top (\mathbf{y} - \bar{\mathbf{y}}) . \quad (4)$$

The “if” direction of this theorem is a standard result by direct calculation, while the “only if” direction was recently proved by the present author [Cho25]. The Bregman divergence is a measure of the difference between the value of ϕ at $\mathbf{y}$ and the value of the linearization (first-order Taylor expansion) of ϕ around $\bar{\mathbf{y}}$ evaluated at $\mathbf{y}$. The Bregman divergence can be interpreted as a measure of dissimilarity of $\mathbf{y}$ from $\bar{\mathbf{y}}$. It is always non-negative and is zero if and only if $\mathbf{y} = \bar{\mathbf{y}}$. The Bregman divergence is not a metric, as it is not symmetric and does not satisfy the triangle inequality, although it does satisfy many other useful properties [BMDG05]. Examples of Bregman divergences include the squared Euclidean distance $g(\mathbf{y}, \bar{\mathbf{y}}) = \|\mathbf{y} - \bar{\mathbf{y}}\|_2^2$ (corresponding to $\phi(\mathbf{y}) = \frac{1}{2}\|\mathbf{y}\|_2^2$) and the Kullback-Leibler divergence $g(\mathbf{y}, \bar{\mathbf{y}}) = \sum_{j=1}^\ell y_j \log(y_j/\bar{y}_j)$ (corresponding to $\phi(\mathbf{y}) = \sum_{j=1}^\ell y_j \log y_j$). A geometric view of Equation (3) is given in Figure 3.

A property of Bregman divergences which will be useful later on is that they are *locally quadratic*, in the sense

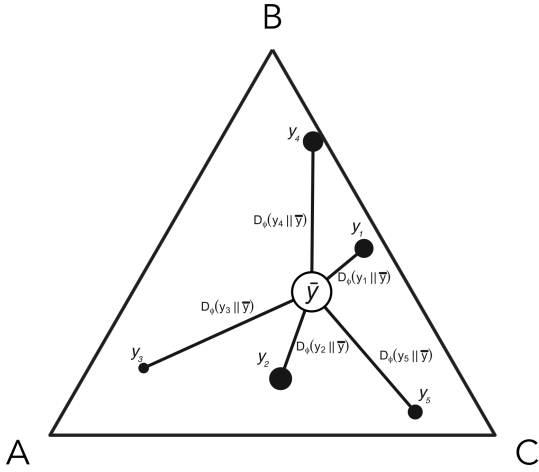


Figure 3: Illustration of Equation (3). Points y_i , $i = 1, \dots, 5$ are sized according to their weight μ_i and are located in the simplex Δ_3 according to their demographic distribution. The Jensen information $I_\phi(\mathcal{P})$ is the weighted mean of the Bregman divergences $D_\phi(y_i \parallel \bar{y})$ from each point to the mean $\bar{y}$. The visual intuition relies on the viewing the Bregman divergence as “distance-like,” though it should be remembered that it is not a metric.

that for any $\mathbf{y} \in \text{Int}(\Delta_\ell)$ and δ in the tangent space to Δ_ℓ at $\mathbf{y}$, we have

$$D_\phi(\mathbf{y} + \delta \parallel \mathbf{y}) = \frac{1}{2} \delta^\top \nabla^2 \phi(\mathbf{y}) \delta + o(\|\delta\|_2^2), \quad (5)$$

where $\nabla^2 \phi(\mathbf{y})$ is the Hessian of ϕ at $\mathbf{y}$ and $o(\|\delta\|_2^2)$ is a term which is asymptotically smaller than $\|\delta\|_2^2$ as $\|\delta\|_2 \rightarrow 0$. Importantly, since ϕ is strictly convex, the Hessian $\nabla^2 \phi(\mathbf{y})$ is positive definite for all $\mathbf{y} \in \text{Int}(\Delta_\ell)$, and so the leading term in the expansion is a positive definite quadratic form. This in turn provides another sense in which Bregman divergences are distance-like, behaving locally like squared Mahalanobis distances with respect to the Hessian of ϕ .

3.1 Properties of the Jensen Information as a Segregation Measure

In an influential article, Reardon and Firebaugh [RF02] propose a set of desirable criteria for multigroup segregation measures, and assess a number of existing measures against these criteria with proofs and counterexamples.⁴ Of the six measures evaluated by the authors, exactly two of these are Jensen informations, corresponding to scaled versions of the Shannon entropy ψ_H and the Simpson index ψ_S . These two measurements come out on top in terms of satisfying the various criteria proposed by the authors. In particular, Reardon and Firebaugh “...conclude that the [Shannon mutual information] is the most conceptually and mathematically satisfactory index...”. It turns out this isn’t a coincidence: many desirable intuitions about segregation come “for free” from convexity.

Some of the Reardon and Firebaugh criteria are trivially satisfied by any Jensen information; for example, their *size invariance* criterion expresses simply the desire that f be a function only of the joint population *distribution* and not absolute population counts. Here we’ll state one of the nontrivial properties and give sufficient criteria for a Jensen information to satisfy it. The intuition expressed by the *exchanges* property is that when individuals move from locations where they are relatively over-represented to locations where they are relatively under-represented, the population becomes less segregated. Let $\tilde{e}_j \in \mathbb{R}^\ell$ denote a standard basis vector in $\mathbb{R}^\ell$.

⁴In the segregation measurement setting the mutual information is sometimes referred to as the *Theil index* or the *information theory index*.

Definition 7. An exchange of magnitude ϵ of groups a and b between locations i and j is a map $\text{exc}_{a,b,i \leftrightarrow j}^{(\epsilon)} : \mathcal{P}_{nd\ell} \rightarrow \mathcal{P}_{nd\ell}$ which takes a population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu)$ and returns a new population $\mathcal{P}' = (\mathcal{X}, \mathbf{Y}', \mu')$ such that

$$\begin{aligned}\mu' &= \mu, \\ \mathbf{y}'_i &= \mathbf{y}_i + \frac{\epsilon}{\mu_i}(\tilde{\mathbf{e}}_b - \tilde{\mathbf{e}}_a), \\ \mathbf{y}'_j &= \mathbf{y}_j + \frac{\epsilon}{\mu_j}(\tilde{\mathbf{e}}_a - \tilde{\mathbf{e}}_b).\end{aligned}$$

Definition 8. A segregation measurement S satisfies the *exchanges property of Reardon and Firebaugh* [RF02] if whenever $y_{ia} < y_{ja}$ and $y_{ib} > y_{jb}$, we have $S(\text{exc}_{a,b,i \leftrightarrow j}^{(\epsilon)}(\mathcal{P})) < S(\mathcal{P})$ for all ϵ sufficiently small.

Theorem 3.2. If ϕ is separable⁵ and strictly convex, then the Jensen information I_ϕ satisfies the exchanges property of Reardon and Firebaugh.⁶

The two Jensen informations in Reardon and Firebaugh’s study are also the only two segregation measures to satisfy what they call “additive organizational decomposability,” which expresses predictable behavior at multiple levels of spatial aggregation. This property is expressed by eq. (7) in Section 4.2.

4 Spatial Analysis with the Jensen Information

Somewhat figuratively, the problem of “spatializing” a segregation measure is the problem of distinguishing the checkered segregation pattern of Figure 1(c) and the monolithic segregation pattern of Figure 1(d). There are several intuitions that can help us distinguish the two patterns:

⁵It is possible to weaken the separability requirement slightly; it is sufficient for ϕ to have coordinatewise monotone gradients, for which it is in turn sufficient that the Hessian $\nabla^2 \phi$ be strictly diagonally dominant.

⁶Reardon and Firebaugh [RF02] also propose a *transfers property*, a stronger form of the exchanges property, and assert that the Shannon mutual information satisfies this property. This author is unable to follow the proof and has performed computational experiments suggesting the transfers property does not indeed hold for the Shannon mutual information.

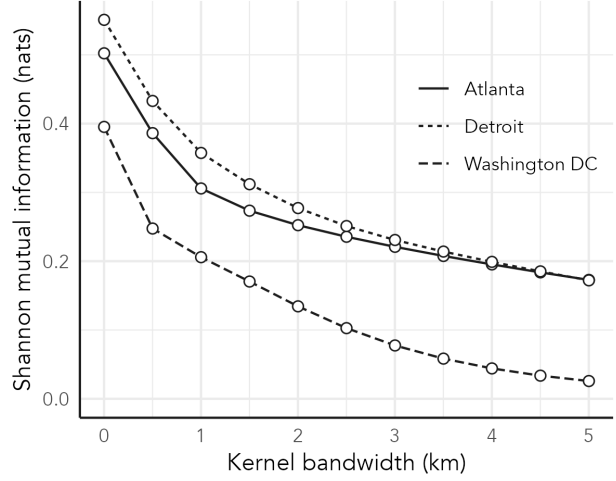


Figure 4: Illustration of spatial profile curves for segregation under Gaussian Radial Basis (RBF) smoothing as the bandwidth is varied. Cities display spatially “coarser” segregation when the value of the Jensen information remains close to its maximum value even as the smoothing bandwidth is increased.

- **Boundaries:** The checkered pattern displays more *boundaries* between regions of different demographic composition.
- **Scale:** The checkered pattern has a smaller *spatial scale* of segregation, in the sense that it is easier to spatially partition the population into homogeneous subpopulations.
- **Perturbation:** If we imagine adding random noise by randomly moving individuals to nearby locations, the checkered pattern will tend to become integrated more rapidly than the monolithic pattern.

These intuitions are all entangled, but lead to different directions in terms of measurement.

4.1 A Perturbative View: Spatial Smoothing

One approach to spatialization, now standard in the quantitative sociology literature [RO04], is to preprocess

the population data by applying a spatial smoothing kernel. To construct a spatialized measure of segregation that can address the checkerboard problem, we simply replace the original population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu)$ with a smoothed population $\mathcal{P}' = (\mathcal{X}, \mathbf{Y}', \mu)$, where $\mathbf{Y}'$ is obtained by applying a smoothing kernel to the rows of $\mathbf{Y}$: for each location i ,

$$\mathbf{y}'_i = \frac{1}{Z_i} \sum_{j=1}^n k(\mathbf{x}_i, \mathbf{x}_j) \mathbf{y}_j ,$$

where $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty)$ is a smoothing kernel and $Z_i = \sum_{j=1}^n k(\mathbf{x}_i, \mathbf{x}_j)$ is a normalization constant. Possible choices of the smoothing kernel k include parametric kernels such as the Gaussian radial basis function $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 / 2\sigma^2)$, or data driven kernels based on road network distance or commute time between locations [Rob18]. After applying the smoothing kernel, any Jensen information may be used to measure segregation in the smoothed population $\mathcal{P}'$. In their theoretical analysis of spatialized segregation measures, Reardon and O'Sullivan [RO04] find that spatialized measures corresponding to Shannon and Simpson informations, again the only Jensen informations amongst their considered measures, satisfy the most of their proposed criteria for spatialized segregation measures, largely by inheriting the favorable properties the general class of Jensen informations.

4.2 A Scale-Based View: Spatial Information Decomposition

Another response to the MAUP is to consider the question of how a segregation measure should change as we modify the areal units under study. We cannot change the areal units arbitrarily, but we can *aggregate* them into larger units. The behavior of a segregation measure under aggregation can be used to describe the scale of segregation in a population. Returning to the grid-cities in Figure 1, for example, the aspatial Shannon mutual information of both panels (c) and (d) is equal to $\log 2$. If we draw a vertical, north-south line down the center of each city, however, aggregating each into two east-west mega-regions, their behaviors contrast: since the two mega-regions in panel (c) have identical demographic compositions, the aspatial Shannon mutual information of the

aggregated population drops to 0, while in panel (d), the two mega-regions are still perfectly segregated, and the aspatial Shannon mutual information remains $\log 2$. It is intuitive that in the case of panel (c), the segregation no longer visible *between* mega-regions should now be seen *within* the mega-regions. This intuition comes “for free” with Jensen information measures via Equation (7) below.

Definition 9 (Spatial Aggregation). Let $n' < n$ and let $\pi : [n] \rightarrow [n']$ be a surjective map. A *spatial aggregation* of a population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$ by π is a new population $\mathcal{P}' = (\mathcal{X}', \mathbf{Y}', \mu') \in \mathcal{P}_{n'd'\ell}$ such that

$$\mu'_j = \sum_{i \in \pi^{-1}(j)} \mu_i, \quad \mathbf{y}'_j = \frac{1}{\mu'_j} \sum_{i \in \pi^{-1}(j)} \mu_i \mathbf{y}_i, \quad \mathbf{X} \in \mathbb{R}^{n' \times d}.$$

The definition of $\mathbf{X}'$ is left intentionally left flexible; one possible choice is to set $\mathbf{x}'_j = \frac{1}{\mu'_j} \sum_{i \in \pi^{-1}(j)} \mu_i \mathbf{x}_i$, the μ -weighted centroid of the locations in the j th aggregated unit.

Definition 10 (Conditional Jensen Information). Let $\mathcal{P}'$ be a spatial aggregation of $\mathcal{P}$ by π . The *conditional Jensen information* of $\mathcal{P}$ given $\mathcal{P}'$ is

$$I_\psi(\mathcal{P} \mid \mathcal{P}') = \sum_{j=1}^{n'} \mu'_j \underbrace{\left(\phi(\mathbf{y}'_j) - \frac{1}{\mu'_j} \sum_{i \in \pi^{-1}(j)} \mu_i \phi(\mathbf{y}_i) \right)}_{I_\psi(\mathcal{P}_j)}, \quad (6)$$

where $\mathcal{P}_j$ is the subpopulation of $\mathcal{P}$ corresponding to the j th aggregated unit. The conditional Jensen information is the mean Jensen information of the subpopulations defined by the aggregated units, weighted by the population of each aggregated unit.

It follows that $I_\psi(\mathcal{P} \mid \mathcal{P}') \geq 0$ for all populations $\mathcal{P}$ and spatial aggregations $\mathcal{P}'$, with equality if and only if the demographic distribution is uniform within each aggregated unit. A direct calculation shows

$$I_\psi(\mathcal{P}) = I_\psi(\mathcal{P}') + I_\psi(\mathcal{P} \mid \mathcal{P}'). \quad (7)$$

Equation (7) is a form of decomposition into “between aggregated unit” information (first term) and “within aggregated unit” information (second term). Since both

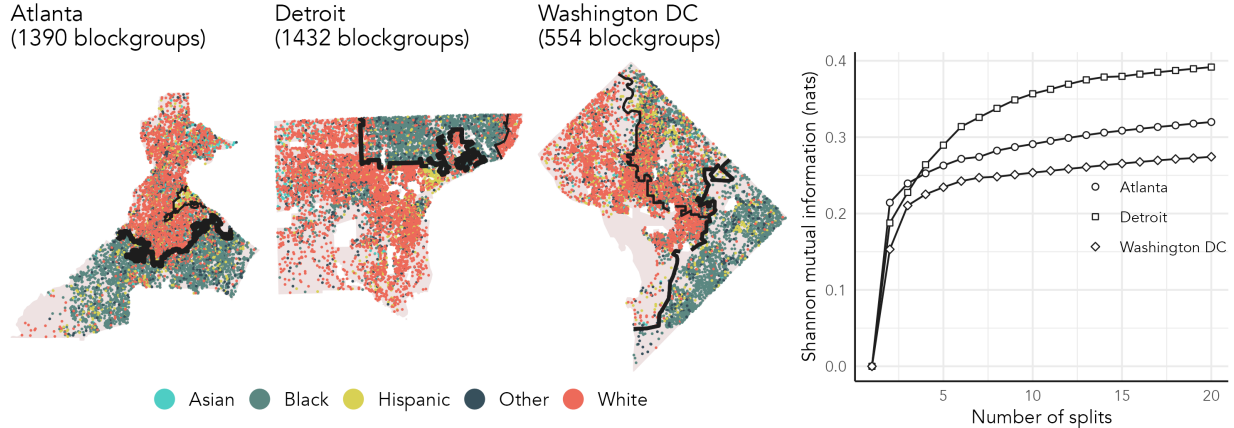


Figure 5: Learned major boundaries (heavy black lines) and information loss associated to them for each of three sample cities. In each map, dots represent 1,000 individuals of the specified demographic category.

terms are nonnegative, it follows that aggregation cannot increase the Jensen information.

Spatial information aggregation is not itself a complete “solution” to MAUP. First, we remain at the mercy of the highest-resolution data available. Second and more fundamentally, the choice of aggregation π is *itself* arbitrary, often based on nested administrative boundaries such as Census block groups, tracts, and counties. One approach to obtaining non-arbitrary aggregations is to seek aggregations which maximize the first, between-unit segregation term in eq. (7) while minimizing the second. The task of forming such spatial aggregations is often called *regionalization*; [Kir22] offers a helpful overview as well as a novel method. One approach, suggested by the present author, is simply to greedily combine adjacent areal units which least decrease the between-unit segregation term in eq. (7) [Cho17]. The resulting hierarchical clustering algorithm has the interpretive benefit that boundaries between aggregated areal units can be assigned information content associated with their contribution to the between-unit segregation term in eq. (7).

4.3 A Boundary-Based View: Local Information

A third approach to spatializing segregation measures follows the intuition that the checkerboard pattern of

Figure 1(c) has more *boundaries* or *transitions* between demographically contrasting regions than the monolithic pattern of Figure 1(d). In this setting, it is convenient to model the observed demographic distribution via a differentiable function $\Upsilon : \mathcal{R} \rightarrow \Delta_\ell$, where $\mathcal{R} \subset \mathbb{R}^d$ represents continuous geographic space and $\mathbf{y} = \Upsilon(\mathbf{x})$ gives the demographic distribution at location $\mathbf{x} \in \mathcal{R}$. This modeling assumption is arguably unrealistic given the presence of sharp boundaries in real-world settled environments, but can be enforced through the application of a differentiable smoothing kernel to the demographic data as described in Section 4.1. Under this set of assumptions, the tendency of the demographic distribution to change with space is given by the Jacobian of Υ , denoted $\nabla \Upsilon(\mathbf{x}) \in \mathbb{R}^{\ell \times d}$. One could in principle describe the “boundariness” of a point $\mathbf{x} \in \text{relint}(\mathcal{R})$ via a norm of the Jacobian such as the Frobenius norm: $b(\mathbf{x}) = \|\nabla \Upsilon(\mathbf{x})\|_F^2 = \sum_{i=1}^{\ell} \sum_{j=1}^d (\partial \Upsilon_i / \partial x_j)^2$. An aggregate measure of boundariness over the entire region $\mathcal{R}$ is then given by the integral $B(\mathcal{R}) = \int_{\mathcal{R}} b(\mathbf{x}) d\mathbf{x}$. This straightforward approach has an important limitation: it is insensitive to the choice of diversity function ϕ and Jensen information I_ϕ used in analysis.

We can obtain a generalization of this approach by considering the behavior of the Jensen information I_ϕ within a small neighborhood about a point $\mathbf{x} \in \text{relint}(\mathcal{R})$. As illustrated in Figure 7, a neighborhood of

a point in the interior of a monolithic region will appear homogeneous and therefore have a “local” Jensen information near 0, while a point near a boundary between contrasting regions will appear heterogeneous and therefore have a higher value of the local information. To make these intuitions precise, we define a localized version of the Jensen information for the continuous setting. For the remainder of this section, we define a continuous Jensen information as

$$I_\phi(\Upsilon, \mathcal{R}) = \int_{\mathcal{R}} (\phi(\Upsilon(\mathbf{x})) - \bar{\mathbf{y}}(\mathcal{R})) d\mu(\mathbf{x}) , \quad (8)$$

where $\bar{\mathbf{y}}(\mathcal{R}) = \int_{\mathcal{R}} \Upsilon(\mathbf{x}) d\mu(\mathbf{x})$ is the mean demographic distribution over the region $\mathcal{R}$.

Definition 11. Let $\Upsilon : \mathcal{R} \rightarrow \Delta_\ell$ be a differentiable function representing the demographic distribution over a region $\mathcal{R} \subset \mathbb{R}^d$, and let $\phi : \Delta_\ell \rightarrow \mathbb{R}$ be a strictly convex function. The *local information* of Υ at a point $\mathbf{x} \in \text{relint}(\mathcal{R})$ is

$$j_\phi(\Upsilon, \mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{I_\phi(\Upsilon, B_\epsilon(\mathbf{x}))}{\epsilon^2} , \quad (9)$$

where $B_\epsilon(\mathbf{x}) = \{\mathbf{y} \in \mathcal{R} : \|\mathbf{y} - \mathbf{x}\|_2 < \epsilon\}$ is the open ball of radius ϵ centered at $\mathbf{x}$.

We can either calculate the local information at many points for use in visualization, or integrate the local information over the entire region to obtain an aggregate spatialized measure, the *mean local information*:

$$J_\phi(\Upsilon, \mathcal{R}) = \int_{\mathcal{R}} j_\phi(\Upsilon, \mathbf{x}) d\mu(\mathbf{x}) . \quad (10)$$

To do this, however, we must know that the limit in eq. (9) exists and must be able to approximate it in a computationally tractable way.

Theorem 4.1 ([Cho17]). *Fix $\mathbf{x} \in \mathcal{R}$ and let $\mathbf{y} = \Upsilon(\mathbf{x})$. The limit in eq. (9) exists and its value is*

$$j_\phi(\Upsilon, \mathbf{x}) = \frac{1}{8\pi} \text{trace} \left(\nabla \Upsilon(\mathbf{x})^\top \nabla^2 \phi(\mathbf{y}) \nabla \Upsilon(\mathbf{x}) \right) . \quad (11)$$

The proof [Cho17] relies on the local quadratic property of Bregman divergences eq. (5).

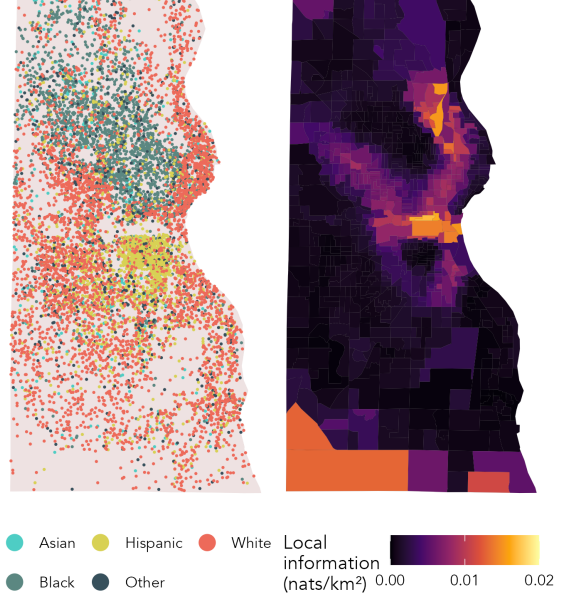


Figure 6: Local information in Milwaukee, Wisconsin. A smoothing kernel of bandwidth 1 km was applied to the original population data, and the local information was computed at each pixel using eq. (12).

In the common case that ϕ is separable, this expression reduces to

$$j_\phi(\Upsilon, \mathbf{x}) = \frac{1}{8\pi} \sum_{j=1}^{\ell} \sum_{i=1}^d \frac{\partial^2 \phi}{\partial y_j^2} \left(\frac{\partial \Upsilon_j}{\partial x_i} \right)^2 . \quad (12)$$

This expression can be viewed as a weighted Frobenius norm with weights given by the entries of the Hessian of the diversity function ϕ . The local information therefore captures the dependence of a spatial measure on *both* the spatial distribution of the population Υ and the choice of diversity function ϕ . To compute the local information in practice, we must approximate the spatial derivatives of Υ . This author previously performed this task via smoothing followed by local least-squares approximation [Cho17], but other estimation methods may prove more effective in practice.

Extensions to Ordinal Segregation Measures

One of the aims of this article has been to abstract attention away from specific measures of diversity and segregation and towards the properties and methods that we can obtain using only minimal additional assumptions. The payoff is that, to apply the theory and methods described, one need only choose a diversity function ϕ and check that it is strictly convex (and, in some cases, separable). Consider, for example, the demographics of *income* and income segregation. In this setting, we might care less about *diversity* with respect to income categories⁷ and more about (approximate) *equality*. There are at least two approaches which fit naturally into the framework developed above.

In one, we define the diversity function ϕ in terms of the cumulative distribution function of wealth. Let $p(w)$ give the proportion of individuals in spatial unit i with wealth w , which we assume to be an integer. Define the vector giving the probability mass function $\mathbf{p} = (p(1), p(2), \dots, p_{w_{\max}})^\top \in \mathbb{R}^{w_{\max}}$, as well as the vector $\mathbf{c} = (p(1), p(1) + p(2), \sum_{i=1}^{w_{\max}} p(i))^\top$. We then have $\mathbf{c} = \mathbf{S}\mathbf{p}$, where $\mathbf{S}$ is the lower-triangular matrix whose lower triangle consists of ones. We let $\phi(\mathbf{p}) = \|\mathbf{c}\|_2^2 = \|\mathbf{S}\mathbf{p}\|_2^2$. Since ϕ is the composition of a full-rank linear map and a strictly convex function, ϕ is itself strictly convex. The corresponding Bregman divergence is the ℓ_2 distance between CDFs, $D_\phi(\mathbf{p} \parallel \bar{\mathbf{p}}) = \|\mathbf{c} - \bar{\mathbf{c}}\|_2^2$. In this formulation, ϕ is nonseparable.

An alternative is to change the role of *people* from units of population to “categories” themselves. A population displaying perfect equality is one in which a given unit of wealth has equal chance of belonging to any given person; that is, in which the distribution of wealth is uniform over all such “categories.” The Shannon entropy of the distribution of wealth over individuals then provides a measure of equality; it is maximized at complete equality and minimized when a single individual possesses all available wealth.⁸ The resulting measure is separable and the analysis tools described in the previous sections apply directly. A limitation of this approach, however,

is that since populations (and therefore numbers of “categories”) differ, direct comparison of segregation levels between urban areas may pose interpretive challenges.

How Many Measurements?

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⁷Does the existence of billionaires in a population contribute to a valuable sense of “income diversity?”

⁸This approach is similar to that of Theil [The67]

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A Supplementary Material

A.1 Pielou’s Third Diversity Axiom

The last of Pielou’s three axioms relates to the behavior of a diversity measure when multiple, independent demographic attributes are considered simultaneously.

Additive under independent attributes. “In a multiple classification (two or more qualitative variates) where the classifications are independent, the total diversity should equal the sum of the respective diversities.” This statement can be formalized as follows, interpreting “independent” in the sense of probability distributions: suppose that, for a given population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$, there exist two other populations $\mathcal{P}_1 = (\mathcal{X}, \mathbf{U} \in \mathbb{R}^{n \times \ell_1}, \mu)$ and $\mathcal{P}_2 = (\mathcal{X}, \mathbf{V} \in \mathbb{R}^{n \times \ell_2}, \mu)$ such that $\ell = \ell_1 \ell_2$ and a bijection $\alpha : [\ell_1] \times [\ell_2] \rightarrow [\ell]$ such that $y_{ij} = u_{i\alpha_1(j)} v_{i\alpha_2(j)}$ for all $i \in [n]$ and $j \in [\ell]$. Then $f(\mathcal{P}) = f(\mathcal{P}_1) + f(\mathcal{P}_2)$.

This axiom, in combination with the other two and under the mild assumption of separability and continuity, is sufficient to uniquely characterize the Shannon entropy:

Theorem A.1. *If a function $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is strictly concave, continuous⁹, separable, and satisfies the additivity*

⁹Convex functions are continuous on the interior of their domain, so this stipulation requires only that they are also continuous at the boundary.

*under independent attributes property, then*¹⁰

$$\psi(\mathbf{y}) = -c \sum_{j=1}^{\ell} y_j \log y_j, \quad (13)$$

for some constant $c > 0$.

A proof is given is given by Chakrabarti and Chakrabarty [CC05], where additivity under independent attributes corresponds to the authors’ Axiom 4.¹¹

B Supplementary Proofs

Proof of Theorem 2.1. Let ϕ be strictly convex. We’ll first assume that ϕ is nonnegative. Then, for any $I \subset [\ell]$ and $\lambda = \frac{|I|}{\ell}$, we have

$$\begin{aligned} \phi\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) &= \phi\left(\lambda \frac{1}{|I|} \mathbf{e}_I + (1-\lambda) \frac{1}{\ell-|I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &< \lambda \phi\left(\frac{1}{|I|} \mathbf{e}_I\right) + (1-\lambda) \phi\left(\frac{1}{\ell-|I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &\leq \phi\left(\frac{1}{|I|} \mathbf{e}_I\right), \end{aligned}$$

where the first inequality is by convexity and the second is by nonnegativity.

Now, relax the requirement that ϕ is nonnegative. Since the simplex Δ_ℓ is compact (closed and bounded) and ϕ is continuous, ϕ attains a minimum on Δ_ℓ , which we denote $\phi_{\min} = \min_{\mathbf{y} \in \Delta_\ell} \phi(\mathbf{y})$. Then, let $\phi'(\mathbf{y}) = \phi(\mathbf{y}) - \phi_{\min}$, which is nonnegative and convex. The calculation above now goes through replacing ϕ with ϕ' , proving that $\phi'\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) \leq \phi'\left(\frac{1}{|I|} \mathbf{e}_I\right)$ and therefore that $\phi\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) \leq \phi\left(\frac{1}{|I|} \mathbf{e}_I\right)$. $\square$

Proof of Theorem 2.2. To demonstrate maximization under the uniform, for any vector $\mathbf{y} \in \Delta_\ell$,

$$\mathbf{e}_{[\ell]}/\ell = \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \sigma(\mathbf{y}),$$

¹⁰Throughout this article, we will use the notation $\log$ to denote the natural logarithm, as well as the standard convention that $0 \log 0 = 0$.

¹¹The authors’ statement of their result also hypothesizes nonnegativity of ϕ , but this hypothesis is not required by their proof.

where $\mathcal{S}$ is the set of all permutations of $[\ell]$. It follows that

$$\begin{aligned}\psi\left(\frac{1}{\ell}\mathbf{e}_{[\ell]}\right) &= \psi\left(\frac{1}{|\mathcal{S}|}\sum_{\sigma \in \mathcal{S}}\sigma(\mathbf{y})\right) \\ &< \frac{1}{|\mathcal{S}|}\sum_{\sigma \in \mathcal{S}}\psi(\sigma(\mathbf{y})) = \psi(\mathbf{y}),\end{aligned}$$

where the inequality is by an inductive application of strict concavity and the final equality is by symmetry. $\square$

Proof of Theorem 3.2. The proof is by direct calculation. Let $\mathcal{P}' = \text{exc}_{a,b,i \leftrightarrow j}^{(e)}(\mathcal{P})$, and let $\mathbf{y}'_i$ and $\mathbf{y}'_j$ be the demographic vectors at locations i and j in $\mathcal{P}'$. Since the global demographic vector $\bar{\mathbf{y}}$ is unchanged by the exchange, we have

$$\begin{aligned}I_\psi(\mathcal{P}) - I_\psi(\mathcal{P}') &= \sum_{i=1}^n [\mu_i \psi(\mathbf{y}_i) - \mu'_i \psi(\mathbf{y}'_i)] \\ &= \mu_i [\psi(\mathbf{y}_i) - \psi(\mathbf{y}'_i)] + \mu_j [\psi(\mathbf{y}_j) - \psi(\mathbf{y}'_j)] \\ &= \mu_i \nabla \psi(\mathbf{y}_i)^\top (\mathbf{y}_i - \mathbf{y}'_i) \\ &\quad + \mu_j \nabla \psi(\mathbf{y}_j)^\top (\mathbf{y}_j - \mathbf{y}'_j) \\ &\quad + o(\|\mathbf{y}_i - \mathbf{y}'_i\|) + o(\|\mathbf{y}_j - \mathbf{y}'_j\|).\end{aligned}$$

since $\mathbf{y}_k = \mathbf{y}'_k$ for all $k \notin \{i, j\}$ and $\mu_k = \mu'_k$ for all k . Using now that $\mathbf{y}_i - \mathbf{y}'_i = \lambda_i(\tilde{\mathbf{e}}_a - \tilde{\mathbf{e}}_b)$ and $\mathbf{y}_j - \mathbf{y}'_j = \lambda_j(\tilde{\mathbf{e}}_b - \tilde{\mathbf{e}}_a)$ and ignoring the higher-order terms, we have

$$I_\psi(\mathcal{P}) - I_\psi(\mathcal{P}') \simeq \mu_i \lambda_i \nabla \psi(\mathbf{y}_i)^\top (\tilde{\mathbf{e}}_a - \tilde{\mathbf{e}}_b) + \mu_j \lambda_j \nabla \psi(\mathbf{y}_j)^\top (\tilde{\mathbf{e}}_b - \tilde{\mathbf{e}}_a).$$

By the separability assumption, we have $\nabla \psi(\mathbf{y}) = (\phi'(y_1), \dots, \phi'(y_\ell))^\top$, and so

$$\begin{aligned}I_\psi(\mathcal{P}) - I_\psi(\mathcal{P}') &\simeq \mu_i \lambda_i (\phi'(y_{ia}) - \phi'(y_{ib})) + \mu_j \lambda_j (\phi'(y_{jb}) - \phi'(y_{ja})) \\ &= \epsilon(\phi'(y_{ia}) - \phi'(y_{ib}) + \phi'(y_{jb}) - \phi'(y_{ja})) \\ &= \epsilon(\phi'(y_{ia}) - \phi'(y_{ja}) + \phi'(y_{jb}) - \phi'(y_{ib})) \\ &> 0.\end{aligned}$$

The last line follows from the assumptions that $y_{ia} < y_{ja}$ and $y_{ib} > y_{jb}$, and the fact that, if ϕ is strictly convex, then ϕ' is strictly increasing, and so $\phi'(y_{ia}) - \phi'(y_{ja}) > 0$ and $\phi'(y_{jb}) - \phi'(y_{ib}) > 0$. Choosing ϵ^* sufficiently small to justify the linear approximation completes the proof. $\square$

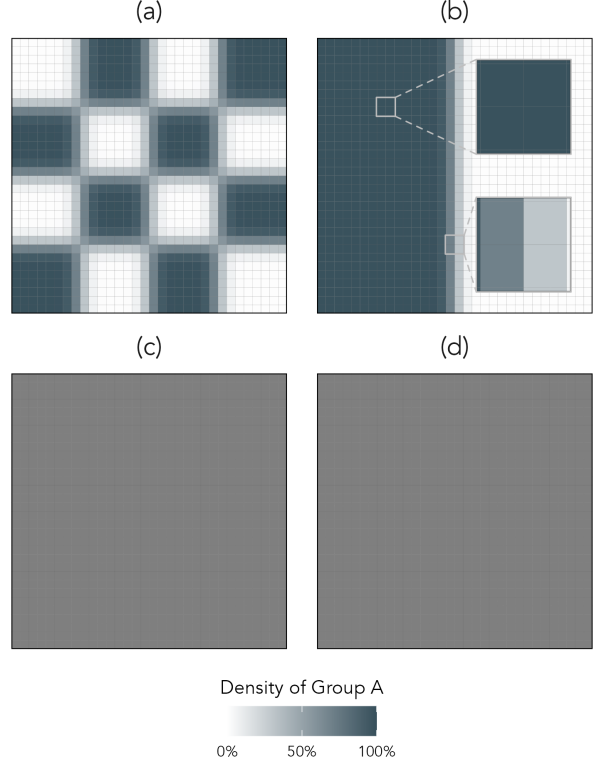


Figure 7: The smoothing bandwidth is 1 pixels. The mean local information of the checkerboard pattern is NaN , while the mean local information of the segregated pattern is NaN .

C Supplementary Figures

D Supplementary Exercises

Exercise D.1. Give a heuristic justification for the convention that $0 \log 0 = 0$ by proving that $\lim_{x \rightarrow 0^+} x \log x = 0$.